



**SKY
TECH**

• A SKY AGRO COMPANY •

BREWERY AND BEVERAGE EQUIPMENT
CATALOGUE

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About SKY TECH



OUR ORIGINS

SKY AGRO LTD began as an agricultural trading company, exporting oilseeds and grains from Ukraine. Malting barley was part of that portfolio, which led us to breweries, and breweries led us to everything else.

We established SKY MALT to supply British brewers with quality malt at competitive prices. Spending time in breweries, talking to the people running them, we kept hearing the same thing: equipment was either out of reach on price, or not backed by anyone who could provide proper support when something went wrong.

In late 2025, we launched SKY TECH to change that. Great equipment, properly supported, accessible to independent breweries.

PRODUCTION

Our equipment is built to order: food-grade stainless steel, Siemens control electronics, and components specified by us and assembled in China by partners we know well.

Our engineering team works with clients from the start, reviewing specifications and layouts, making sure what gets built is right for the brewery it is going into. Every order is accompanied by full engineering documentation, including layout drawings, P&ID diagrams, and 3D renders, so there are no surprises when equipment arrives on site.

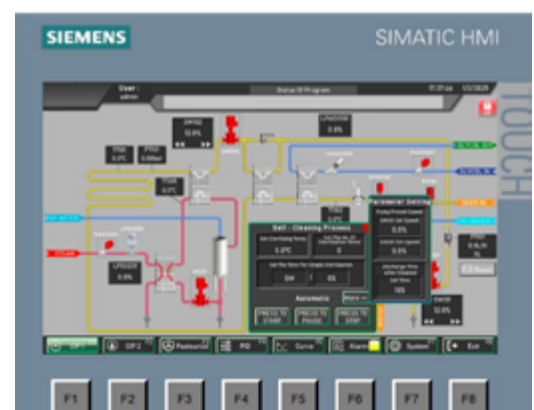
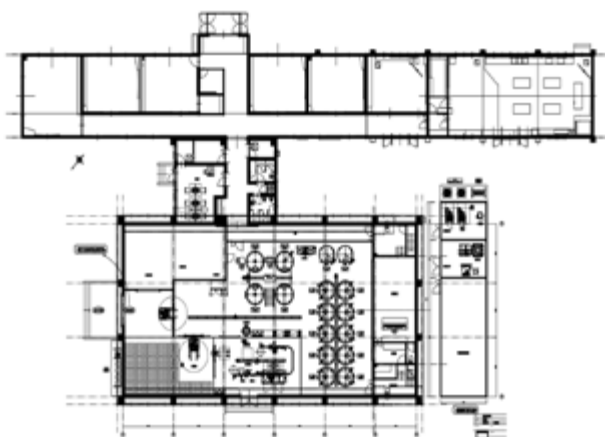
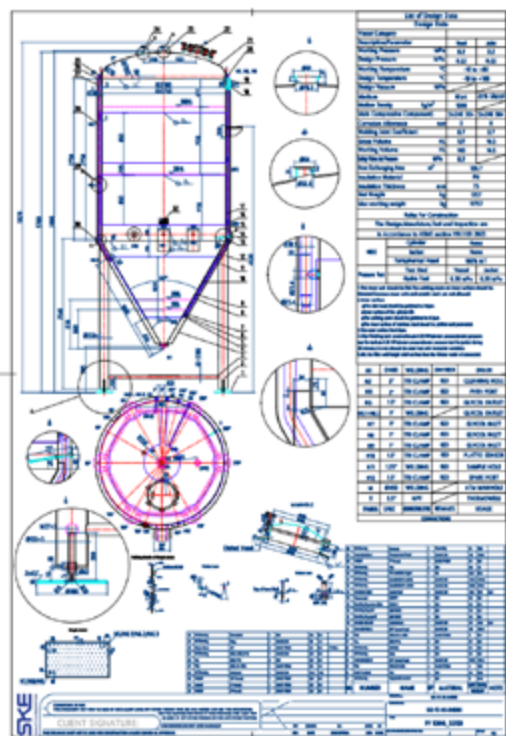
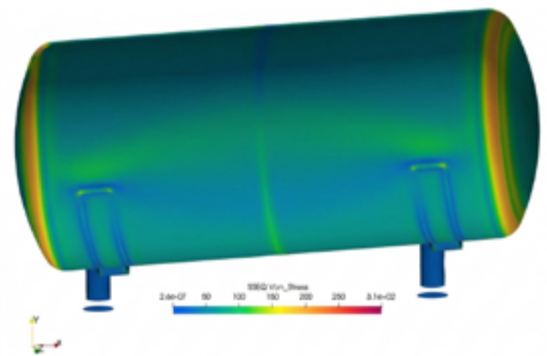
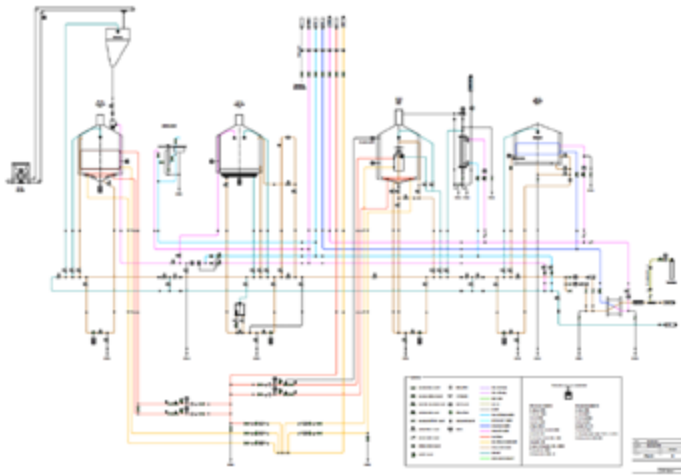
INSTALLATION AND SUPPORT

Before shipment, our equipment goes through a Factory Acceptance Test: assembled, commissioned, and checked against spec at the factory.

After the equipment arrives, our UK-based team covers installation support, technical troubleshooting, and ongoing monitoring.

We work with breweries at every stage: those setting up for the first time, and established operations looking to expand or upgrade. The scale varies, but our approach does not.

SKY TECH sits alongside SKY MALT and SKY LABS as part of a broader offer for breweries: ingredients, chemicals, and brewing equipment from one place.



Service and Project Delivery

We are committed to providing not only high-quality equipment, but end-to-end service: technical planning, installation, and post-installation support. Most of our equipment is built to order, and we are able to provide turnkey solutions for most anything brewery-related, including planning and designing breweries from the ground up.

Every project, regardless of scale, follows the same five-stage process.

1

Customisation and System Design

A dedicated project manager and engineering team are assigned at project start. Deliverables include P&ID drawings, project layout, and 2D/3D fabrication drawings.

2

Manufacturing and Assembly

Equipment is fabricated to confirmed drawings. Weekly production reports are issued throughout. All vessels and systems undergo a Factory Acceptance Test (FAT) prior to shipment.

3

Logistics

Shipment is managed by a dedicated logistics manager. Skid-mounted systems are pre-assembled and tested for straightforward site connection.

4

Installation and On-site Testing

On-site commissioning covers all equipment, control systems, and instrumentation. A water-run test simulates process conditions before product is introduced.

5

Post-installation Service and Warranty

All equipment carries a one-year warranty. Engineering documentation is retained for the life of the installation, with support available for any issues arising after handover.



Standards and Certifications

All SKY TECH equipment is manufactured and certified to the standards required for brewery and beverage installations across the UK, Europe, North America, and Australia.

Pressure Vessels

Pressure vessels, including fermentation vessels and brite beer tanks, are engineered to ASME standards and available with PED (EU Directive 2014/68/EU) and AS 1210 certification.

Control Systems and Electrical components

Control systems and electrical components are certified to CE, UL, and CUL standards. All enclosed industrial control panels are built and wired to the electrical standards of the destination market, with full documentation supplied at handover.

Factory Acceptance Testing

Every vessel and system undergoes a Factory Acceptance Test prior to shipment: assembled, commissioned, and inspected against specification at the factory. Pressure vessels are tested for leakage, and surface finish is verified against the specified roughness standard.

Documentation

Every order is supplied with a full documentation package: material certificates, inspection reports, engineering drawings, and quality documentation. For electrical systems, wiring schematics and programming manuals are included as standard. Everything is supplied at handover, before commissioning begins.



Brewhouses

2 hL – 100 hL

Brewhouse systems encompass the mashing, lautering, wort boiling, and whirlpool vessels at the centre of every brewing process. Available in two-vessel and three-vessel configurations, built for brewpubs, nanobreweries, and microbreweries.

PILOT

Pilot Brewing System

<5 hL

SKID-MOUNTED

Brewhaus Senior

15 hL – 100 hL

SKID-MOUNTED

Brewhaus Pro

2 hL – 15 hL

CUSTOM

Turnkey Solutions

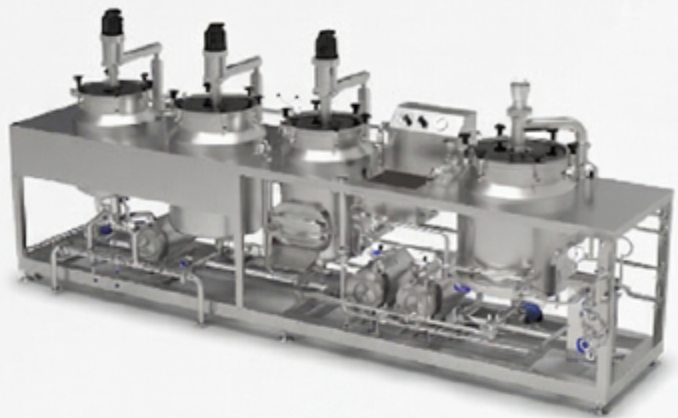
Custom



Pilot Brewing System

Compact 4-vessel skid-mounted pilot brewhouse, 2–15 hL, for recipe development and process testing.

More information on p. 10-11



Brewhaus Pro

Skid-mounted brewhouse for smaller breweries, <5 hL, steam/electric/direct fire heating.

More information on p. 12-13



Brewhaus Senior

Skid-mounted brewhouse for medium to large craft breweries, 15–100 hL, 3V/4V/5V configurations, fully automated PLC/HMI.

More information on p. 14-15





Brewhouses

Pilot Brewing System

Overview

The Pilot Brewing System is a compact, skid-mounted 4-vessel brewhouse designed for recipe development, process testing, and quality control at laboratory or pilot scale. It is a functionally complete smaller-scale brewhouse incorporating the same process vessel sequence and control architecture as the full-scale production systems, allowing recipes and process parameters developed on the pilot system to be transferred directly to the production brewhouse.

The system is steam-heated and supports step-mashing and decoction processes. PLC and HMI control is provided as standard, with manual valves for process flexibility. A full self-cleaning function covers hot water rinse and caustic/sanitiser CIP without manual disassembly. Whirlpool hopping and kettle souring capability are included as standard.

All vessels, pipework, pumps, a plate heat exchanger, and instrumentation are pre-assembled on an integrated skid platform with no dead legs in the pipework or vessels.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity range	<5 hL (custom options available)
Shell material	SS304 food grade
Configuration	4-vessel: mash cooker, mash tun, lauter tun, kettle and whirlpool tank
Heating method	Steam
Control	PLC/HMI with manual valves
Self-cleaning	Yes; hot water rinse and caustic/sanitiser CIP; no disassembly required
Installation	Skid-mounted; plug-and-play
Application	Recipe development, process testing, and quality control



Brewhouses

Pilot Brewing System

Key Features

VESSEL CONFIGURATION

4-vessel configuration: mash cooker, mash tun, lauter tun, and kettle and whirlpool tank

Whirlpool hopping and kettle souring capability as standard

CONTROL AND CLEANING

PLC and HMI control with manual valves for process flexibility

Self-cleaning function: hot water rinse and caustic/sanitiser CIP; no manual disassembly required

INSTALLATION AND INTEGRATION

Skid-mounted; all vessels, pipework, pumps, plate heat exchanger, and instrumentation pre-assembled on a single platform

No dead legs in pipework or vessels

HEATING AND PROCESS

Steam heating; step-mashing and decoction capability



Brewhouses

Brewhaus Pro

2 hL – 15 hL

Overview

The Brewhaus Pro is a skid-mounted brewhouse system manufactured in SS304 stainless steel and designed for brewpubs, nanobreweries, and microbreweries. The system is available from 2 hL to 15 hL batch capacity and integrates all process vessels, pipework, pumps, a plate heat exchanger, and instrumentation onto a single skid platform for simplified on-site installation.

Heating is available by steam, electric, or direct fire. Control is manual or semi-automatic; the semi-automatic configuration supports both manual and automatic operating modes, allowing the brewer to switch between them during a brew session. The system is certified to CE, UL/CUL, or AS standards depending on market.

The Brewhaus Pro is designed for a broad range of beer styles. The kettle and whirlpool tank (KWT) supports whirlpool hopping, and the system includes kettle souring capability as standard. A self-cleaning function covers hot water rinse and caustic/sanitiser clean-in-place (CIP) for the entire mashing process without manual disassembly.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity range	2–15 hL
Shell material	SS304 food grade
Internal finish	2B, Ra ≤ 0.6 µm
External finish	#4 polished
Heating method	Steam, electric, or direct fire
False-bottom open area	>18%
Control	Manual or semi-automatic
Self-cleaning	Yes; hot water rinse and caustic/sanitiser CIP; no disassembly required
Installation	Skid-mounted; plug-and-play



Brewhouses

Brewhaus Pro

Key Features

CAPACITY AND CONFIGURATION

Available from 2 hL to 15 hL batch capacity

Custom vessel combinations available

CONTROL AND CLEANING

Manual or semi-automatic control; international-brand instrumentation throughout

Self-cleaning function for the full mashing circuit; no manual disassembly required

INSTALLATION AND INTEGRATION

Skid-mounted, all process equipment pre-assembled. Plug-and-play installation

No dead legs in pipework or vessels

HEATING AND PROCESS

Heating by steam, electric, or direct fire

KWT head space $\geq 70\%$; internal wall polished to $Ra \leq 0.2 \mu m$; IFM anti-overflow sensor

Whirlpool hopping and kettle souring capability as standard

System Components

CONTROL SYSTEM

Manual or semi-automatic configurations

Manual: temperature regulation, VFD speed control, flow/volume measurement, emergency stop, anti-overflow protection

Semi-automatic: automated flow, temperature, and volume control across all vessels

VFD control on all pumps, rake drive, mill, and conveyor

Alarms: emergency stop, kettle anti-overflow, low compressed-air pressure

KETTLE AND WHIRLPOOL TANK (KWT)

Headspace $\geq 70\%$; internal surfaces $Ra \leq 0.2 \mu m$, passivated

Integrated boiling and whirlpool functions

IFM anti-overflow sensor

MASH LAUTER TUN

False bottom: SS milled, open area $>18\%$

VFD rake control; knife position 5–10 mm above false bottom

Differential pressure sight glasses for grain bed monitoring

Backflush devices for false bottom cleaning between batches

Emergency stop at spent grain door

Water-mixing station for mash-in volume and temperature control



Brewhouses

Brewhaus Senior

15 hL – 100 hL

Overview

The Brewhaus Senior is a skid-mounted brewhouse system manufactured in SS304 stainless steel, designed for medium to large-scale craft breweries operating at 15 hL to 100 hL batch capacity. The system supports 4–8 brewing batches per day and is available in 3-vessel, 4-vessel, and 5-vessel configurations to suit varying process requirements and production throughputs.

The Brewhaus Senior places emphasis on process adaptability, energy recovery, operational automation, and brewing safety. High-level automation via a customised PLC and human-machine interface (HMI) programme reduces labour input and batch-to-batch variability. All principal instrumentation uses international-brand components; all mechanical parts use high-quality domestic-brand components.

The system integrates all process vessels, pipework, pumps, a plate heat exchanger, and instrumentation onto a modular brew deck. Piping diameters are calculated for target flow rates at each process stage, with valve positions set for ergonomic access. A full self-cleaning function covers the mashing circuit without manual disassembly.

Technical Specifications

PARAMETER	SPECIFICATION
Batch capacity	15–100 hL
Daily throughput	4–8 batches per day
Shell material	SS304 food grade
Internal finish	2B, Ra ≤ 0.6 µm
External finish	#4 polished
Vessel configurations	3-vessel (3V); 4-vessel (4V); 5-vessel (5V)
Heating method	Steam (standard); other options available on request
Whirlpool base slope	2%
Control	PLC/HMI; fully automatic; customised per installation
Self-cleaning	Yes; hot water rinse and caustic/sanitiser CIP; no disassembly required
Installation	Skid-mounted; plug-and-play



Brewhouses

Brewhaus Senior

Key Features

CAPACITY AND CONFIGURATION

15–100 hL batch capacity; 3V, 4V, and 5V configurations

Supports 4–8 brewing batches per day

5V configuration: decoction and step-mashing; optional boiling buffer tank for maximum daily throughput

CONTROL AND CLEANING

Customised PLC/HMI control programme per installation; full automation of flow, temperature, volume, and motor speed

Self-cleaning function for the full mashing circuit; no manual disassembly required.

INSTALLATION AND INTEGRATION

Modular brew deck; all pipework, pumps, plate heat exchanger, and instrumentation pre-installed

No dead legs in pipework or vessels.

HEATING AND PROCESS

Lauter tun with VFD rake speed control; false bottom opening rate and diameter calculated for target filtration efficiency

Whirlpool tank with tangential inlet, 2% sloped base, and three height-adjustable wort outlets

System Components

MASHING SYSTEM

Starch conversion and flavour extraction

Water-mixing station for mash-in volume and temperature control

Steam heating; step-mashing capable

BOILING KETTLE

Controlled boiling and evaporation for stable wort quality

Anti-overflow sensor

BREW DECK AND PIPEWORK

SS304 modular brew deck; integrates all pipework, pumps, heat exchangers, and sensors

Pipe sizing calculated for stable transfer; layout eliminates dead spots

LAUTERING SYSTEM

False bottom, rake geometry, and vessel dimensions optimised to prevent compaction

VFD rake control; variable speed across filtration stages

WHIRLPOOL TANK

Tangential inlet; 2% sloped base; optimised geometry for trub separation

Height-adjustable outlets for staged wort discharge

Cellar Tanks

Fermentation & Conditioning

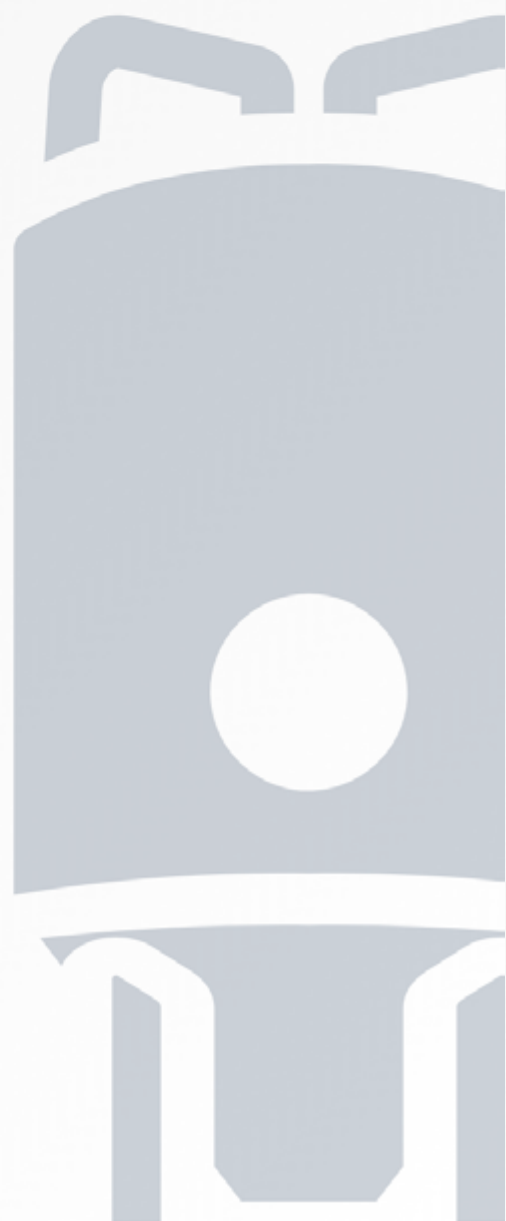
Cellar tanks hold and condition beer from the end of fermentation through to packaging. SKY TECH fermentation vessels and brite beer tanks are manufactured in food-grade SS304 stainless steel, certified to PED, ASME, and AS1210 standards, and available from 1 hL to 500 hL.

FERMENTATION

Fermentation Vessels 1 hL – 300 hL

CONDITIONING

Brite Beer Tanks (BBTs) 1 hL – 500 hL



Fermentation Vessels (CCVs)

SS304 cylindroconical fermentation vessels, 1–300 hL, PED/ASME/AS1210 certified, indoor and outdoor configurations available.

More information on p. 18-19



Brite Beer Tanks (BBTs)

SS304 conditioning vessels, 1–500 hL, available in vertical, horizontal, and stacked configurations.

More information on p. 20-21





Cellar Tanks

Fermentation Vessels

1 hL – 300 hL

Overview

Cylindroconical fermentation vessels (CCVs) are manufactured in food-grade SS304 stainless steel, sized from 1 hL pilot batches to 300 hL production runs. A 60° conical bottom promotes efficient yeast sedimentation and collection. Dimple-jacketed cooling on the cylinder body and cone bottom, combined with 3" PU foam insulation, gives the brewer precise temperature control through primary fermentation, conditioning, and cold crashing.

All vessels are certified to PED (EU 2014/68/EU), ASME, and AS1210 as standard. Factory-finished to 2B internal finish ($Ra \leq 0.6 \mu m$), delivered ready for installation. Indoor and outdoor configurations share the same specification; no additional outer casing is required for outdoor use.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity range	1–300 hL
Shell material	SS304 food grade
Internal finish	2B, $Ra \leq 0.6 \mu m$
External finish	#4 polished
Head type	Top dished head
Cone angle	60°
Insulation	3" PU foam throughout
Cooling jacket	Dimple jacket on cylinder
Working pressure	0.15–0.3 bar; pressure vessel rated
Welding	Full protection-gas atmosphere welding
Leg configuration	Four adjustable legs; anti-seismic feet optional
Installation	Indoor or outdoor; stackable available

Vessel Types

TYPE	APPLICATION
CCV – Indoor	Standard production fermenter. Insulated and jacketed for controlled indoor environments. Stackable where headroom allows.
CCV – Outdoor	Same specification as indoor CCV. Suitable where indoor floor space is constrained or capacity is expanding without structural work.
Open-Top	Traditional open vessel for top-fermenting ales and cask production. Open fermentation character integral to process.
Yorkshire Square	Two-chamber vessel for classic Northern English ales. The upper chamber promotes yeast head formation and top-cropping, suited to traditional ale strains that benefit from yeast-rich fermentation character.



Cellar Tanks

Fermentation Vessels

Key Features

CONSTRUCTION AND MATERIALS

SS304 food-grade shell; 2B internal finish ($R_a \leq 0.6 \mu\text{m}$), #4 polished external finish

Full protection-gas atmosphere welding; all welds internally ground flush

60° cone angle for efficient yeast sedimentation and complete drainage

Top dished head; top manway or side manhole specified at order

TEMPERATURE CONTROL

Dimple jacket on cylinder body and cone bottom for glycol-based cooling

3" PU foam insulation for thermal stability during extended conditioning

Side-mounted thermowell (sanitary grade) for temperature monitoring at the vessel wall

CIP AND HYGIENE

Sanitary-grade CIP rotary spray ball for full internal coverage without dead zones

Side CIP arm with bung valve and pressure gauge; all fittings are sanitary grade

Sample valve (sanitary grade) for in-process quality checks

Racking arm in cone for controlled transfer at end of fermentation

PRESSURE AND SAFETY

Pressure/vacuum relief valve supplied as standard; vessels are pressure vessel rated

Working pressure specified at order to applicable market standards

Standard and Optional Fittings

STANDARD FITTINGS

Pressure/vacuum relief valve

CIP rotary spray ball (sanitary grade)

Side CIP arm with bung valve and pressure gauge

Thermowell (sanitary grade, side-mounted)

Sample valve (sanitary grade)

Racking arm in cone (sanitary grade)

Top manway or side manhole (specified at order)

Dry hopping port

Bottom discharge

OPTIONAL FITTINGS

In-pressure dry hopping device

Tank-top device

Anti-seismic feet

Stacking configuration

Larger-capacity vessels (>300 hl, assembled on-site)



Cellar Tanks

Brite Beer Tanks

1 hL – 500 hL

Overview

Brite beer tanks (BBTs) are manufactured in food-grade SS304 stainless steel and are designed to serve the full range of production scales, from 1 hL pilot operations to 500 hL commercial installations. All principal process contact surfaces are finished to 2B standard at $Ra \leq 0.6 \mu m$ internally and #4 polished externally.

All vessels are built to ASME standards and supplied with PED (EU Directive 2014/68/EU) and AS1210 certification as standard, ensuring compliance across regulated markets worldwide. With 3-inch PU foam insulation and dimple-jacketed cooling on the cylinder, BBTs deliver consistent temperature control throughout conditioning and carbonation.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity range	1–500 hL
Shell material	SS304 food grade
Internal finish	2B, $Ra \leq 0.6 \mu m$
External finish	#4 polished
Head type	Top dished head
Insulation	3" PU foam throughout
Cooling jacket	Dimple jacket on cylinder
Working pressure	Available on request; pressure vessel rated
Configurations	Vertical, horizontal, or stacked
Leg configuration	Four adjustable legs; anti-seismic feet optional
Installation	Stackable available

Vessel Types

TYPE	APPLICATION
Vertical	Standard brite beer tank for most production environments. Insulated and jacketed, suited to conditioning and serving from a single vessel in a controlled cellar setting.
Horizontal	An alternative orientation suited to low-headroom environments or where floor layout makes horizontal vessels preferable. Identical specification to the vertical configuration.
Stacked	Configured for vertical stacking to maximise conditioning capacity within a limited floor footprint. Suited to expanding bright beer capacity without extending the building.
Single-wall serving tank	A non-insulated option for short-term serving applications, typically in bar or taproom environments where ambient temperature control is managed externally.



Cellar Tanks

Brite Beer Tanks

Key Features

CONSTRUCTION AND MATERIALS

SS304 food-grade shell; 2B internal finish ($Ra \leq 0.6 \mu m$), #4 polished external finish

Full protection-gas atmosphere welding; all welds internally ground flush.

Top dished head; top manway or side manhole specified at order.

CIP AND HYGIENE

Sanitary-grade CIP rotary spray ball for full internal coverage without dead zones.

Side CIP arm with bung valve and pressure gauge; all fittings are sanitary grade.

Sample valve (sanitary grade) for in-process quality checks.

TEMPERATURE CONTROL

Dimple jacket on cylinder body for glycol-based cooling.

3" PU foam insulation for thermal stability during conditioning and carbonation.

Side-mounted thermowell (sanitary grade) for temperature monitoring at the vessel wall.

PRESSURE AND SAFETY

Pressure/vacuum relief valve supplied as standard; vessels are pressure vessel rated.

Working pressure specified at order to applicable market standards.

Standard and Optional Fittings

STANDARD FITTINGS

Pressure/vacuum relief valve

CIP rotary spray ball (sanitary grade)

Side CIP arm with bung valve and pressure gauge

Thermowell (sanitary grade, side-mounted)

Sample valve (sanitary grade)

Top manway or side manhole (specified at order)

Bottom discharge

OPTIONAL FITTINGS

Hot water heating jacket
(in-lieu of or in addition to cooling jacket)

Anti-seismic feet

Stacking configuration (for optimal space usage)

Horizontal orientation

Single-wall (non-insulated) configuration for serving applications

CIP (Cleaning-in-Place)

Single vessel – multi-vessel

Clean-in-place systems circulate cleaning and sanitising solutions through vessels and pipework without disassembly, maintaining hygiene standards between production runs.

Available as a portable cart for flexible single-vessel use or as a fixed automated station for multi-vessel installations.

MOBILE
CIP Cart

FIXED
Auto CIP Station



CIP Cart

Portable manual CIP unit for cleaning fermenters, BBTs, and ancillary equipment.

[More information on p. 24-25](#)



Auto CIP Station

Fixed automated CIP station with PLC control, programmable loop selection, and multi-circuit capability.

[More information on p. 26-27](#)





CIP (Cleaning-in-Place)

CIP Cart

Overview

The CIP Cart is a compact, portable clean-in-place unit designed for intuitive, hassle-free operation. Engineered to clean the interior surfaces of pipes, vessels, and associated fittings, it is widely used across the brewing, food, dairy, beverage, and pharmaceutical industries. Its mobile design makes it equally suited to permanent cellar installation and remotely located or temporarily connected equipment.

Technical Specifications

PARAMETER	SPECIFICATION
Tank capacities	Standard: 50 L, 100 L, 200 L caustic tank and sanitiser tank. Custom sizes available upon request.
Temperature control	Digital temperature controller; innovative temperature maintenance of caustic solution during cleaning process
Pump	Variable speed pump (VFD); brands include Danfoss, Invertek, ABB
Sanitiser line	Separate sanitiser line prevents cross-contamination between sanitiser and caustic fluid
Liquid safety	Inline liquid sensor & temperature probe assembly prevents dry-firing of heating elements
Valves	Directional valves for recirculating and transferring to target tanks



CIP (Cleaning-in-Place)

CIP Cart

Key Features

PERFORMANCE AND SAFETY

Repeatable, reliable cleaning for consistent product quality

Solution circulation between target vessel and CIP tank; reduces operator working time

Integrated vortex prevention device; protects pump from air cavitation

Splash-prevention design during solution recovery and return

COMPATIBILITY AND MOBILITY

Compatible with all food and beverage equipment

Fully mobile; suitable for permanent cellar installation or hard-to-reach equipment

CHEMICAL MANAGEMENT

Chemical recapture system; solutions recyclable across multiple cleaning cycles

Heated tanks maintain solution temperature throughout the CIP process

Multiple reservoir configuration; supports pre-filling, pre-heating, and multiple solution types simultaneously



CIP (Cleaning-in-Place) Auto CIP Station

Overview

The Auto CIP Station is a modular, fully automated clean-in-place system designed for targeted cleaning of pipework, tanks, filling machines, and other production equipment. Cleaning parameters, including time, temperature, pH level, pressure, and flow rate, are fully adjustable to meet the specific requirements of each application.

The system is supplied as a plug-and-play unit: pre-assembled with pipelines, plate heat exchanger, pumps, and instrumentation, and fully tested prior to delivery. All wetted components are manufactured from stainless steel with heat-resistant seals, and the system is fully compliant with brewing, beverage, and food industry regulations.

Technical Specifications

The Auto CIP Station is custom-designed to your process requirements. The following components form the standard configuration, with the number and type of tanks adjusted to match the specific CIP programme required.

PARAMETER	SPECIFICATION
Tanks	Water/Rinse Tank, Steriliser Tank, Caustic Tank, Acid Tank, CIP Recovery Tank. Material: SS316L or SS304. Quantity customised to CIP process.
Pipelines and Manifold	Automatic butterfly valves; supply and return valve manifold or switch panel with feedback signal. Material: SS316L or SS304.
Pumps	CIP supply pumps and CIP return pumps. Material: SS316L or SS304. VFD on supply pump.
Chemical Dosing	Dosing pump and chemical drum.
Heating System	Plate heat exchanger; steam valves, trap, and filter.
Instrumentation	RTD (supply & return), flow sensor (supply), conductivity sensor (return), pH sensor (supply), tank level sensors, pressure sensors (CIP supply & steam supply).
Control System	PLC + HMI, full automatic. Remote monitoring.



CIP (Cleaning-in-Place) Auto CIP Station

Key Features

PERFORMANCE AND SAFETY

Repeatable, reliable cleaning for consistent product quality

Solution circulation between target vessel and CIP tank; reduces operator working time

Integrated vortex prevention device; protects pump from air cavitation

Splash-prevention design during solution recovery and return

COMPATIBILITY AND MOBILITY

Compatible with all food and beverage equipment

Fixed installation; pre-assembled and fully tested prior to delivery as a plug-and-play unit.

CHEMICAL MANAGEMENT

Chemical recapture system; solutions recyclable across multiple cleaning cycles

Heated tanks maintain solution temperature throughout the CIP process

Multiple reservoir configuration; supports pre-filling, pre-heating, and multiple solution types simultaneously

Pasteurisation

Batch & inline

Pasteurisation extends product shelf life by applying controlled heat treatment to eliminate spoilage organisms.

Available in two formats: batch treatment of bottled and canned product, and continuous inline treatment prior to packaging.

BATCH	INLINE
Cabinet Pasteuriser	Flash Pasteuriser



Cabinet Pasteuriser

Batch tunnel pasteuriser for bottled and canned beer, using steam at up to 100°C.

More information on p. 30-31



Flash Pasteuriser

Inline HTST pasteuriser for beer, RTDs, and carbonated beverages; 85–90% energy recovery via PHX loop.

More information on p. 32-33





Pasteurisation

Cabinet Pasteuriser

Overview

Our Cabinet Pasteurisers are manufactured in SS304 stainless steel and use a built-in steam boiler to supply steam at up to 100°C within the chamber for batch pasteurisation of bottled and canned beer, milk, and other beverages. Sterilisation temperature and time are fully adjustable and automatically controlled throughout the pasteurisation cycle. The system is suitable for a wide range of product and container types.

Technical Specifications

Our Cabinet Pasteuriser is available in two configurations: single and double-cabin.

PARAMETER	SINGLE-CABIN PASTEURISER	DOUBLE-CABIN PASTEURISER
Capacity	300 × 500 mL or 420 × 330 mL per batch	600 × 500 mL or 840 × 330 mL per batch
Operating temperature range	60–93°C	60–93°C
Max temperature capability	100°C	100°C
Working pressure	1.0 bar	1.0 bar
Heating source	External boiler (customer-supplied); internal electric boiler; internal gas boiler	Internal electric boiler
Power	24 kW	45 kW
Voltage	400 V / 50 Hz / 3Ph (customisable)	400 V / 50 Hz / 3Ph (customisable)
Total dimensions (cabin)	1100 × 1050 × 1870 mm	2100 × 1050 × 1870 mm
Internal cart dimensions	660 × 915 × 1720 mm	660 × 915 × 1720 mm per cart
Approximate weight	600 kg	700 kg



Pasteurisation Cabinet Pasteuriser

Key Features

PASTEURISATION AND TEMPERATURE CONTROL

Accurate control of pasteurisation temperature and time

360° steam surround sterilisation for heat uniformity throughout the chamber

Dual temperature sensors to monitor and verify pasteurisation temperature

Cooling water supply to bring product down to ambient temperature after pasteurisation

SAFETY

Maximum working pressure 1.0 bar; no risk of bottle burst

Automatic water supply protection and anti-dry heating protection

DESIGN AND USABILITY

Sanitary, compact module; stable operation and low maintenance

Unique stainless steel mobile pallet truck for straightforward loading and unloading

Process Description

Total cycle time is approximately 1 hour per batch, comprising three phases:

- 1 Heating Phase**
Temperature ramp from ambient to 70°C+, achieved within 30–40 minutes.
- 2 Holding Phase**
Temperature maintained at pasteurisation setpoint for 5–10 minutes.
- 3 Cooling Phase**
Temperature reduction from 70°C+ back to ambient, achieved within 10–20 minutes.



Pasteurisation

Flash Pasteuriser

Overview

The Flash Pasteurizer is an inline (HTST) pasteurisation system specifically engineered for craft breweries and the beverage industry, suitable for beer, soda, seltzer, RTDs, and other carbonated drinks. Unlike batch cabinet pasteurisation, flash pasteurisation treats product continuously as it flows through the system prior to packaging.

The upgraded design features hard piping configurations for enhanced hygiene and a compact skid structure. A cascade control loop improves temperature control accuracy and minimises deviations caused by material agitation. Highly flexible programming enables curve recording, data output, and batch settings for seamless production data tracking.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity	1000 L/h
Beer-in temperature	1–4°C
Sterilisation temperature	72°C
Holding time	30 seconds
Beer-out temperature	0–4°C
Steam pressure required	>2 bar
Steam consumption	20–25 kg/h
Compressed air required	>5 bar
Glycol flow rate	2.5 t/h
Electrical power	4 kW
Energy recovery	>85%
Control	SIEMENS PLC + 7" SIEMENS HMI, fully automatic



Pasteurisation

Flash Pasteuriser

Key Features

PROCESS AND TEMPERATURE CONTROL

Precise PU setting and automatic control; compatible with variable flow rates and temperature adjustment

Automated cascade control loop for smooth, gentle processing and maximum deviation management

User-friendly interface with flexible pasteurisation range settings

HYGIENE AND CIP

Hygienic pipeline design; minimises dead legs; self-draining capable

Excellent CIP compatibility and hygienic design throughout

INSTALLATION AND OPERATION

Compact skid design; small footprint; plug-and-play installation

High degree of automation with stable operational performance

ENERGY RECOVERY

PHX and loop design recovers up to 85–90% of energy; reduces load on heating and cooling systems

Process Description

The system operates as a three-zone plate heat exchanger, recovering heat from pasteurised product to pre-heat incoming cold product before final heating and cooling. The entire process operates under continuous supervision. Any deviation in sterilisation temperature or pressure triggers an alarm and automatically diverts product into a recirculation loop.

1 Pre-heating

Cold beer (4°C) enters the second zone of the PHX via a boosting pump, where it exchanges heat with the outgoing pasteurised beer (72°C+), pre-heating to approximately 62°C.

2 Heating and Holding

Pre-heated beer is boosted again and enters the third (heating) zone, where it is brought to the sterilisation temperature of 72°C via automated medium temperature regulation. It then passes through the holding pipe and is maintained at this temperature for 30 seconds to achieve sterilisation.

3 Cooling

The pasteurised beer (72°C+) returns to the second zone of the PHX, transferring heat to the incoming cold beer and cooling to approximately 17°C. It then enters the first zone where -1°C glycol water cools it to 4°C for packaging.

Blending and Carbonation

Up to 5.5 g/L CO₂

Inline carbonation and blending systems introduce CO₂ into product and combine multiple liquid streams in a continuous process.

Available in manual, semi-automatic, and fully automatic configurations, including a multi-effect production line with integrated pasteurisation.

BATCH

Manual Carbonation and Blending System

FIXED

Semi-Auto Carbonation and Blending System

BATCH

Auto Carbonation and Blending System

FIXED

Multi-Effect Beverage Production Line



Manual Blending and Carbonation System

Inline carbonation unit, manual operation, up to 5.5 g/L CO₂.

[More information on p. 36-37](#)



Semi-Auto Blending and Carbonation System

Inline carbonation unit, automated blending ratio control, manual CO₂ flow, up to 5.5 g/L CO₂.

[More information on p. 38-39](#)



Auto Blending and Carbonation System

Automatic inline blending and carbonation unit with integrated static blender, fully automatic PLC/HMI recipe control, up to two blending streams, up to 5.5 g/L CO₂.

[More information on p. 40-41](#)



Multi-effect Beverage Production Line

Integrated production line combining blending, carbonation, flash pasteurisation, and dispensing; up to four ingredient streams.

[More information on p. 42-43](#)





Blending and Carbonation

Manual Blending and Carbonation System

Overview

The Manual Carbonation and Blending System is a compact, wheel-mounted unit designed for CO₂ infusion and liquid additive blending in beer and carbonated beverages. All wetted components are manufactured in food-grade stainless steel, with a sanitary corrugated static packing element as the core mixing component.

CO₂ injection is achieved via a Venturi injector, which produces rapid, uniform gas dispersion. Flow control for both the product stream and CO₂ supply is manual, with dedicated flow metres for additive metering enabling precise control of additive-to-base ratios. The packing element is clean-in-place (CIP) compatible with no dead corners.

The system suits small-batch and on-site production applications including beer, sparkling water, soda, and ready-to-drink (RTD) beverages. Its mobile frame and small footprint make it equally suitable for use at fixed cellar stations or temporary production locations.

Technical Specifications

PARAMETER	SPECIFICATION
Shell material	Food-grade stainless steel
CO ₂ injection method	Venturi injector
Mixing element	Sanitary corrugated static packing
Control mode	Manual (flow metres for liquid and CO ₂)
Mobility	Wheel-mounted; no fixed installation required
CIP compatibility	Fully compatible, no dead corners
Application	Multiple options; available on request

Blending and Carbonation

Manual Blending and Carbonation System

Key Features

CARBONATION AND BLENDING

Venturi injector for rapid, uniform CO₂ dispersion into the product stream

Sanitary corrugated static packing for thorough mixing of base liquid and liquid additives

Dedicated flow metres for precise manual control of CO₂ flow, liquid pressure, and additive stream ratios

HYGIENE AND COMPATIBILITY

Fully CIP-compatible; no dead corners

Compatible with beer, sparkling water, soda, and all carbonated beverages

INSTALLATION AND MOBILITY

Wheel-mounted mobile frame; no fixed installation required; repositionable to any production station

Small overall footprint; no ancillary utility connections required beyond CO₂ and liquid supply

Blending and Carbonation

Semi-Auto Blending and Carbonation System

Overview

The Inline Semi-Auto Carbonation and Blending System is a continuous carbonation unit designed for beer and carbonated beverages. CO₂ is introduced inline as product flows through the system, eliminating the downtime associated with batch carbonation. Carbonation level is held within preset standards via precise CO₂ proportion injection, with liquid pressure maintained continuously by an integrated booster pump and pressure-maintaining valve assembly.

The system uses a PIN-POINT injection design: micro-pore stainless steel injector plates disperse CO₂ into ultra-fine 5–9 micron bubbles, maximising gas-liquid contact area and achieving near-100% CO₂ mass transfer efficiency. The design is fully compliant with food and beverage industry hygiene regulations and is clean-in-place (CIP) compatible throughout.

CO₂ flow is manually controlled; base liquid and flavour dosing ratios are automatically regulated. The system suits craft brewery and commercial beverage production environments requiring consistent carbonation at variable flow rates.

Technical Specifications

PARAMETER	SPECIFICATION
Shell material	Food-grade stainless steel
CO ₂ injection method	PIN-POINT micro-pore injector plates; 5-9 micron bubble dispersion
CO ₂ transfer efficiency	Near 100%
Process type	Continuous inline
Control mode	Semi-automatic: automated liquid/flavour ratio; manual CO ₂ and pressure
CIP compatibility	Fully compatible, hygienic design
Capacity	Multiple options; available on request

Blending and Carbonation

Semi-Auto Blending and Carbonation System

Key Features

CARBONATION AND BLENDING

PIN-POINT injection: micro-pore SS injector plates disperse CO₂ at 5–9 micron bubble diameter; near-100% mass transfer efficiency

Integrated booster pump and pressure-maintaining valve assembly for continuous carbonation

Precise automated ratio control for base liquid and flavour dosing; manual control for CO₂ flow and liquid pressure

HYGIENE AND COMPLIANCE

Fully hygienic CIP design; compliant with food and beverage industry regulations

INSTALLATION

Small footprint; no additional utility connections required beyond CO₂ and liquid supply

PROCESS AND OPERATION

Inline continuous process; no batch-related downtime

Blending and Carbonation

Auto Blending and Carbonation System

Overview

The Inline Auto Carbonation and Blending System is a fully automated inline unit for the continuous carbonation and blending of beer, soft drinks, ready-to-drink (RTD) beverages, and other carbonated products. It combines an inline static blending module with a PIN-POINT CO₂ injection system, and is controlled via programmable logic controller (PLC) with a human-machine interface (HMI) for recipe-based operation.

The blending module processes two streams simultaneously: one for base liquid (de-aerated water or beer) and one for syrups, liquid flavours, or alcohol dosing. Each stream is equipped with a feed pump, flow metre, and positioning PID valve, with mixing ratios automatically regulated to preset recipes. Additional streams or powder dosing are available as custom options.

The CO₂/N₂ carbonation line uses a mass flow controller and PID valves for automatic flow regulation, with target carbonation levels set directly in the PLC programme. Maximum carbonation level under standard operating conditions is up to 5.5 g/L. The system is fully CIP-compatible.

Technical Specifications

PARAMETER	SPECIFICATION
Shell material	Food-grade stainless steel
CO ₂ injection method	PIN-POINT micro-pore injector plates; 5-9 micron bubble dispersion
Carbonation range	2–5.5 g/L (standard conditions)
CO ₂ transfer efficiency	Near 100%
Process type	Static inline
Control mode	PLC/HMI; fully automatic; recipe-based
Blending streams	Two standard (base liquid + flavour/syrup/alcohol); additional streams available
Flow control	Feed pump, flow metre, and PID positioning valve per stream
CIP compatibility	Fully compatible, hygienic design
Capacity	Multiple options; available on request

Blending and Carbonation

Auto Blending and Carbonation System

Key Features

CARBONATION AND BLENDING

PIN-POINT micro-pore SS injector plates; 5–9 micron CO₂ bubble dispersion; near-100% mass transfer efficiency

Mass flow controller and PID valves for precise automated CO₂ and N₂ dosing; carbonation range 2–5.5 g/L

Each blending line equipped with feed pump, flow metre, and positioning PID valve for automatic ratio regulation

HYGIENE AND COMPLIANCE

Fully hygienic pipeline design; CIP-compatible throughout

INSTALLATION

Small footprint; compact skid structure suitable for integration into existing production lines

Multiple capacity options available; custom configurations accepted

CONTROL AND AUTOMATION

Fully automatic PLC/HMI control with recipe setting; multiple product programmes storable

Blending and Carbonation Multi-effect Beverage Production Line

Overview

The Multi-effect Beverage Production Line is an integrated, fully inline processing system designed for high-flexibility beverage manufacturing with consistent product quality and minimal product loss. The line combines multi-stream proportional blending, temperature control, inline carbonation, flash pasteurisation, and automated clean-in-place (CIP) in a single continuous process.

De-aerated water (DAW) and up to four additional liquid streams, including wine, spirits, syrup, and flavour, are metered continuously and precisely into a static mixing unit based on operator-selected recipes via the HMI. An optional powder dosing module is available. Each ingredient line is equipped with either a diaphragm or flow metre for flow rate control in accordance with recipe settings.

The system supports both batch and continuous production modes and is suited to alcoholic and non-alcoholic beverages, RTD products, soft drinks, and other carbonated formats. A buffer tank and two storage/dispensing tanks are included as standard, with automated tank switching logic to ensure blending never occurs into a tank currently supplying the filler.

Technical Specifications

PARAMETER	SPECIFICATION
Process stages	Blending, temperature control, carbonation, flash pasteurisation, dispensing
Blending streams	Up to four ingredient streams plus de-aerated water; powder dosing optional
Carbonation range	2–8.5 g/L; CO ₂ or N ₂ ; mass flow controller dosing
Cooling	PHX for crash cooling prior to carbonation
Pasteurisation	Flash pasteuriser; suitable for alcoholic, non-alcoholic, RTD, soft-drink formats
CIP	Semi- or fully automatic; pneumatically controlled valve manifolds; programmable loop selection via HMI
Tank configuration	1 × buffer tank; 2 × storage/dispensing tanks; automated switching logic
Tank pressure (filling)	1.5–2.0 bar
Tank pressure (filler connection)	0.8–1.2 bar
Production modes	Batch or continuous

Blending and Carbonation Multi-effect Beverage Production Line

Key Features

CARBONATION AND BLENDING

Multi-stream inline proportional blending of up to four ingredient streams plus de-aerated water; powder dosing optional

Inline carbonation via mass flow controller; precise CO₂ or N₂ dosing to recipe; carbonation range 2–8.5 g/L with minimal deviation; optional inline dissolving instrument

PHX for rapid temperature control and crash cooling prior to carbonation

HYGIENE AND CONTROL

Automated CIP via pneumatically controlled valve manifolds and programmable loop selection via HMI

Semi- or fully automatic operation; cleaning liquid automatically routed to correct process sections per selected loop

Batch and continuous production modes available

TANK MANAGEMENT

One buffer tank and two storage/dispensing tanks included as standard

Automated tank switching; filler supply tank never simultaneously receiving blended product

Tank pressure regulated via gas-side control: 1.5–2.0 bar during filling, 0.8–1.2 bar prior to filler connection; all adjustable via HMI

PASTEURISATION

Flash pasteurisation module for alcoholic, non-alcoholic, RTD, and soft-drink formats

Dealcoholisation and Deaeration

ABV to < 0.5%

Dealcoholisation and deaeration systems allow breweries to manage alcohol content and dissolved oxygen across their product range.

The dealcoholisation system uses reverse osmosis membrane filtration to reduce beer ABV to below 0.5%; the deaeration system supplies low-oxygen water for dilution, high-gravity brewing, and packaging.

ALCOHOL REMOVAL

Dealcoholisation System

OXYGEN REMOVAL

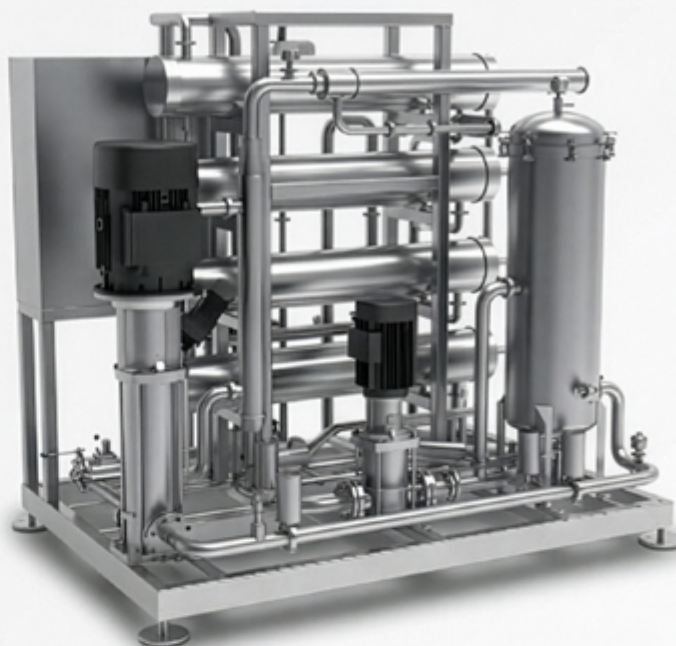
Deaeration System



Dealcoholisation System

Skid-mounted spiral RO membrane unit for reducing beer ABV to below 0.5%, without heat input.

More information on p. 46-47



Deaeration System

CO₂ stripping packed tower system supplying low-oxygen water for dilution, high-gravity brewing, and packaging.

More information on p. 48-49





Dealcoholisation and Deaeration

Dealcoholisation System

Overview

Our Dealcoholisation System is a skid-mounted, spiral reverse osmosis (RO) membrane filtration unit designed for the dealcoholisation of fully fermented beer and other low-alcohol beverages. All principal components are manufactured in food-grade stainless steel. The system operates without heat input, preserving the flavour compounds of the base beer through a gentle, pressure-driven membrane separation process.

Because the membrane process selectively removes ethanol and water while retaining the heavier flavour-active compounds, polyphenols, esters, and other beer aromatics, the alcohol-by-volume (ABV) of the finished product is reduced to below 0.5% whilst the original beer flavour profile is substantially maintained. The diafiltration stage uses de-aerated water to wash out residual alcohol from the concentrated retentate, with the diafiltration volume determining the final ABV of the non-alcoholic (NA) product.

The system is suited to craft and commercial breweries producing non-alcoholic and low-alcohol beer. Capacity is scalable by adding membranes in parallel. Both semi-automatic and fully automatic control configurations are available, and the system integrates two separate clean-in-place (CIP) loops.

Technical Specifications

PARAMETER	SPECIFICATION
Membrane type	Spiral reverse osmosis (RO)
Finished product ABV	<0.5% (typically 0.3-0.5%)
Application	Beer, wine, spirits, and other low-alcohol fermented beverages
Throughput per membrane	100 L/h (additional membranes in parallel for higher capacity)
Operating pressure	Up to 10 bar
CIP loops	Two: one for RO membranes; one for process pipework
Operating modes	Continuous separation; batch separation
Control	Semi-automatic or fully automatic; PLC-based
Cooling	No heat input required; optional inline glycol/chilled liquid cooling heat exchanger
Installation	Skid-mounted



Dealcoholisation and Deaeration Dealcoholisation System

Key Features

MEMBRANE AND CAPACITY

Spiral RO membrane configuration; capacity scalable by adding membranes in parallel; base throughput 100 L/h per membrane

Finished product ABV below 0.5% (typically 0.3–0.5%); variable by pressure, diafiltration volume, and membrane selection

HYGIENE

Two dedicated CIP loops: one for RO membranes, one for process pipework

PROCESS AND CONTROL

Online supervision of permeate, retentate, and diafiltration flow with automatic regulation

Working pressure monitored and interlocked with booster pump

Dual operating modes: continuous and batch separation

TEMPERATURE

No heat input required; optional inline cooling heat exchanger for glycol or chilled liquid temperature control

Process Description

The Dealcoholisation System works in three distinct stages:

1

Concentration

Beer is pressurised to up to 10 bar and circulated through the RO membrane modules. Ethanol and water permeate the membranes and are collected as permeate in the central pipe; this stream exits the system and may be discarded or recovered. The heavier flavour-active compounds are retained in the concentrate (retentate) and recirculated.

2

Diafiltration

Deaerated water is introduced to dilute the concentrated retentate and assist in further alcohol removal. The diafiltration water volume is the primary variable controlling final product ABV. This stage is analogous to a sparging process: repeated membrane passes with fresh water progressively reduce the ethanol content of the recirculating stream.

3

Dilution

Deaerated water is added to restore the product to its original volume and target final ABV before the finished NA beer exits the system.



Dealcoholisation and Deaeration Deaeration System

Overview

The 20 hL per hour de-aerated water system is designed to supply low-oxygen brewing water for applications such as dilution, high-gravity brewing, and packaging. The system uses CO₂ stripping within a packed tower to achieve efficient oxygen removal, ensuring stable water quality and improved product shelf life.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity	20 hL per hour
Tower material	SS304 stainless steel
Tower dimensions	Ø219 mm × 6.7 m
Storage tank volume	4000 / 5000 L
Design pressure	1 bar
Voltage	400 V / 3 Ph / 50 Hz
Pump power	1.5 kW



Dealcoholisation and Deaeration Deaeration System

Key Features

PROCESS AND CONTROL

Efficient CO₂ stripping for oxygen removal

Stable supply with CO₂ back pressure control

Fully automated PLC/HMI control

INSTALLATION

Pre-fabricated pipework for rapid installation

HYGIENE AND CLEANING

Hygienic SS304 construction

Integrated CIP system

System Components

RAW WATER SUPPLY

Flow metre, pneumatic valves, and pre-fabricated pipework

Flow rate controlled via inline measurement; CO₂ dosing matched through HMI-controlled regulation

DEAERATION TOWER

Stainless steel packed tower with water distributor and filling media

Maximises gas-liquid contact and oxygen removal efficiency

CO₂ AERATION SYSTEM

Pressure-controlled CO₂ injection with flow metering

Automated valve control under PLC operation

CONTROL SYSTEM

PLC/HMI-based automated control of flow, level, and CO₂ dosing

Dynamic adjustment based on operating conditions

CIP SYSTEM

Integrated cleaning pipework for automated cleaning cycles

Prevents fouling and maintains hygienic performance

Keg Washing and Filling

10L - 60L kegs

Kegging and keg washing equipment cleans, sanitises, and fills kegs under counter-pressure, maintaining product integrity and minimising oxygen pickup at packaging.

Available from compact two-station washers for smaller operations through to automatic washing and filling lines for higher-volume production.

WASHING	WASHING & FILLING
Keg Washer	Keg Washer and Filler



Keg Washer

Fully automatic two-station keg washing unit, compatible with 10–60 L Euro and US standard kegs, steam or electric heating.

More information on p. 52-53



Keg Washer and Filler

Fully automatic two-station combined keg washing and filling unit, with Siemens PLC/HMI control and CO₂ counter-pressure filling.

More information on p. 54-55



Keg Washing and Filling Keg Washer

Overview

Our Keg Washer is a two-station, fully automatic keg cleaning unit manufactured in SS304 stainless steel, designed for craft breweries operating with Euro or US standard kegs from 10 L to 60 L. Kegs are loaded manually onto the working table; from that point, all cleaning operations are executed via pre-set programmes with no manual connection required. An automatic connector locks the keg spear securely in place for each cycle.

The unit operates two independent cleaning stations simultaneously. Each cycle uses high- and low-pressure pulsed valves to ensure thorough cleaning of the keg stem and internal surfaces. Heating is available by steam or electric element; control is provided by a Siemens programmable logic controller (PLC). The machine is certified to CE, UL, and CUL standards.

The Keg Washer is suited to smaller breweries requiring a compact, reliable, and straightforward keg preparation solution.

Technical Specifications

PARAMETER	SPECIFICATION
Keg compatibility	10–60 L Euro/US standard; spear type configurable to order
Throughput	30–40 kegs/h
Dimensions	1320 × 760 × 1640 mm
Approximate weight	200 kg
Heating method	Steam or electric
Total power	8 kW (electric heating); 2.2 kW (steam heating)
Cleaning tanks	Two 70 L rectangular tanks
Pumps	Two centrifugal cleaning pumps
Valve type	Automatic (pneumatic and solenoid)
Control	Siemens PLC

Keg Washing and Filling Keg Washer

Key Features

CLEANING AND WASHING

Two independent cleaning stations operating simultaneously; high- and low-pressure pulsed valves for thorough cleaning of keg stem and internal surfaces

Two 70 L SS304 rectangular cleaning tanks with two centrifugal cleaning pumps

Pneumatic and solenoid automatic valve control throughout

Heating by steam (2.2 kW) or electric element (8 kW); dedicated temperature control for cleaning solution

CONTROL AND AUTOMATION

Fully automatic operation via Siemens PLC; kegs loaded manually, all subsequent steps via pre-set programme

Automatic connector locks keg spear without manual intervention; reduces operator handling and improves cycle consistency

COMPATIBILITY

Compatible with 10–60 L Euro and US standard kegs; spear type configurable to order

Cleaning Programmes

Two cleaning programmes are available, selected according to available utilities and requirements.

PLAN A: STEAM STERILISATION

Water rinse to remove residuals, caustic clean and rinse, hot water rinse, steam sterilisation, and CO₂ counter-pressure preparation.

PLAN B: CHEMICAL SANITATION

Water rinse to remove residuals, caustic clean and rinse, hot water rinse, chemical sanitisation, and CO₂ counter-pressure preparation.



Keg Washing and Filling

Keg Washer and Filler

Overview

The Keg Washer and Filler is a two-station, fully automatic keg washing and filling unit manufactured in SS304 stainless steel, designed for craft breweries requiring a combined clean-and-fill solution in a single compact machine. Kegs are loaded manually; a double-button start initiates the automated cycle. The machine handles washing, sanitisation, and CO₂ counter-pressure filling in sequence without further operator intervention.

The unit includes a 7-inch colour touchscreen human-machine interface (HMI) and a Siemens programmable logic controller (PLC) for recipe management and process monitoring. Dedicated caustic and sanitiser tanks with a 6 kW low-watt-density immersion heater element maintain solution temperature throughout operation. Two heavy-duty stainless steel pumps provide cleaning and transfer flow. A tuning fork level switch and digital pressure gauge provide process monitoring at the keg connection point.

The Keg Washer and Filler accepts multiple keg sizes and is compatible with Micro Matic keg couplers; spear type is configurable to order. The machine is suited to craft breweries requiring flexible, operator-friendly keg preparation and filling without the footprint or capital cost of a full kegging line.

Technical Specifications

PARAMETER	SPECIFICATION
Keg compatibility	Multiple sizes; Micro Matic couplers; spear type configurable to order
Stations	Two (wash/sanitise + fill)
Operation	Manual keg loading; double-button start; fully automatic cycle
Heater element	6 kW low-watt-density immersion heater
Pumps	Two heavy-duty pumps
Valve type	Low-flow pulse valve for keg stem cleaning
Level monitoring	Tuning fork level switch
Pressure monitoring	Digital pressure gauge
Control	Siemens PLC; 7-inch colour touchscreen HMI

Keg Washing and Filling

Keg Washer and Filler

Key Features

CLEANING AND WASHING

Dedicated caustic and sanitiser tanks; 6 kW low-watt-density immersion heater and temperature probe; level switch for dry-running protection

Two heavy-duty SS304 pumps for cleaning and transfer; low-flow pulse valve for precise keg stem cleaning

CONFIGURATION

Two-station configuration: one station for keg washing and sanitisation, one station for CO₂ counter-pressure filling; operating simultaneously

CONTROL AND AUTOMATION

Fully automatic PLC/HMI operation via 7-inch colour touchscreen; recipe setting, process monitoring, and alarm management from a single interface

Double-button start with keg positioning to confirm correct seating before cycle initiation

Digital pressure gauge and tuning fork level switch for process monitoring at connection point

COMPATIBILITY

Compatible with multiple keg sizes; Micro Matic keg couplers with configurable spear type

Distillation

2 hL – 30 hL

Copper and stainless steel pot still systems for spirits production, with bubble-cap distillation columns, dephlegmator, condenser, and EX-rated control system. Available with gin basket and spirit storage configuration.

POT STILL
Distillation System



Distillation System

Copper and stainless steel pot still for craft spirits production, available as single malt whisky, modern hybrid, and continuous configurations, 2-30 hL capacity.

[More information on p. 58-59](#)





Distillation

Distillation System

Overview

The SKY TECH Distillation System is a copper and stainless steel pot still designed for craft spirits production, including whisky, brandy, gin, and other distilled beverages. The pot is constructed with inner copper and stainless steel cladding, with heating via steam jacket or steam coil; the heating method is specified at order. The system is configured around a bubble-cap distillation column, with the number of plates determining the target alcohol content of the finished spirit.

The system includes a dephlegmator for partial condensation and reflux control, a shell-and-tube condenser, and a spirit parrot for real-time ABV monitoring during the run. A gin basket is available for vapour-path botanical infusion. The control panel is EX-rated throughout, with automatic temperature control across the pot, dephlegmator, and condenser. Three spirit storage tanks for heads, hearts, and tails are included as standard.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity range	2–30 hL
Shell material	Copper inner; stainless steel cladding
Still Types	Single malt whisky, modern hybrid, continuous
Heating method	Steam jacket or steam coil (specified at order)
Column type	Bubble-cap; plate count variable
Spirit storage	Three tanks (heads, hearts, tails)
Gin basket	Optional
ABV monitoring	Spirit parrot (inline, real-time)
Control	EX-rated; automatic temperature control



Distillation

Distillation System

Key Features

POT STILL

Copper inner with stainless steel cladding; steam jacket or steam coil heating, specified at order.

Explosion-proof motor and gearbox, agitator, glass manway, spray balls, temperature and pressure gauges, safety and pressure relief valves.

CONDENSATION AND REFLUX

Tube-type dephlegmator with pneumatic or manual PID modulating valve for automatic or manual temperature control.

Shell-and-tube condenser with pneumatic or manual angle seat valve for automatic or manual temperature control.

OPTIONS

Gin basket for vapour-path botanical infusion.

Custom support frame with integrated heads, hearts, and tails spirit storage tanks.

DISTILLATION COLUMN

Bubble-cap column with red copper bubble caps and sight glass per plate; plate count determines target spirit ABV.

CIP ball and discharge valve per plate.

CONTROL AND SAFETY

EX-rated control panel; automatic temperature control of pot, dephlegmator, and condenser.

Spirit parrot for real-time ABV monitoring between condenser and spirit storage tanks.

Kombucha

Skid-mounted kombucha brewing systems covering the full production process, from tea preparation and primary fermentation through to secondary fermentation, carbonation, and packaging.

BREWING

Kombucha Brewhouse



Kombucha Brewhouse

Skid-mounted three-vessel kombucha brewing system covering tea preparation, primary fermentation, and finishing, with optional agitation mixer.

[More information on p. 62-63](#)





Kombucha

Kombucha Brewhouse

Overview

The Kombucha Brewing System covers the full production process from tea preparation through to secondary fermentation, carbonation, and packaging. The system comprises three vessels, a kombucha brewhouse, fermentation tank, and finishing tank, engineered in food-grade stainless steel and available in a skid-mounted configuration for straightforward installation and expansion.

The brewhouse stage boils water and combines tea and sugar to produce a sweetened tea base, which is cooled via plate heat exchanger before transfer to the fermentation tank. Primary fermentation with a symbiotic culture of bacteria and yeast (SCOBY) takes place over one to three weeks. Secondary fermentation, carbonation, and flavour addition occur in the finishing tank, from which the product can be packaged directly.

Technical Specifications

PARAMETER	SPECIFICATION
Capacity	Custom; available on request
Shell Material	Food-grade stainless steel
Configuration	Three-vessel: brewhouse, fermentation tank, finishing tank
Cooling	Plate heat exchanger
Primary fermentation	1–3 weeks; SCOBY inoculation
Secondary fermentation	In finishing tank; carbonation and flavouring at this stage
Agitation	Optional mixer
Installation	Skid-mounted



Kombucha

Kombucha Brewhouse

Key Features

BREWING AND FERMENTATION

Three-vessel configuration: kombucha brewhouse, fermentation tank, and finishing tank.

Plate heat exchanger for rapid cooling of the tea base prior to fermentation.

Primary fermentation vessel sized for one to three week SCOBY fermentation cycles.

FINISHING AND PACKAGING

Finishing tank supports secondary fermentation, carbonation, and flavour addition in a single vessel.

Product can be packaged directly from the finishing tank; no transfer required.

INSTALLATION AND INTEGRATION

Skid-mounted configuration; modular components for straightforward capacity expansion.

Optional agitation mixer for enhanced process versatility.

Flexible capacity options, specified at order to suit premises size and production volume.

SKY TECH

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